

We have performed the given below stages to create the **Data Visualization**

1. Dataset Download:

Download the College Food Choices dataset in CSV format from the provided project repository. Store the dataset in a dedicated project folder for easy access and management.

2. Data Import:

Import the dataset into Tableau or Microsoft Excel. Ensure all records and fields such as Student ID, Age, Gender, Meal Preference, Dietary Habit, Physical Activity, BMI, and Food Consumption Frequency are imported correctly.

3. Data Inspection:

Examine the dataset to understand its structure, including the number of rows, columns, data types, and available variables. Identify important attributes such as dietary preferences, meal frequency, nutritional intake, health indicators, and lifestyle factors.

4. Data Cleaning:

Remove duplicate entries, handle missing values, correct formatting inconsistencies, standardize categorical values (e.g., Vegetarian, Vegan, Non-Vegetarian), and ensure numerical fields such as BMI, Calories, and Water Intake are stored correctly.

5. Data Validation:

Validate the dataset by checking for incorrect or inconsistent values. Confirm that dietary categories, health metrics, and food consumption records fall within acceptable and realistic ranges.

6. Data Transformation:

Create calculated fields such as Average Daily Calorie Intake, Healthy Eating Score, or BMI Category. Rename columns for clarity, standardize text values, and prepare the dataset for meaningful analysis.

7. Data Filtering:

Filter the data based on relevant criteria such as Gender, Age Group, Diet Type, BMI Category, Meal Frequency, or Physical Activity Level to focus on specific dietary trends and health patterns.

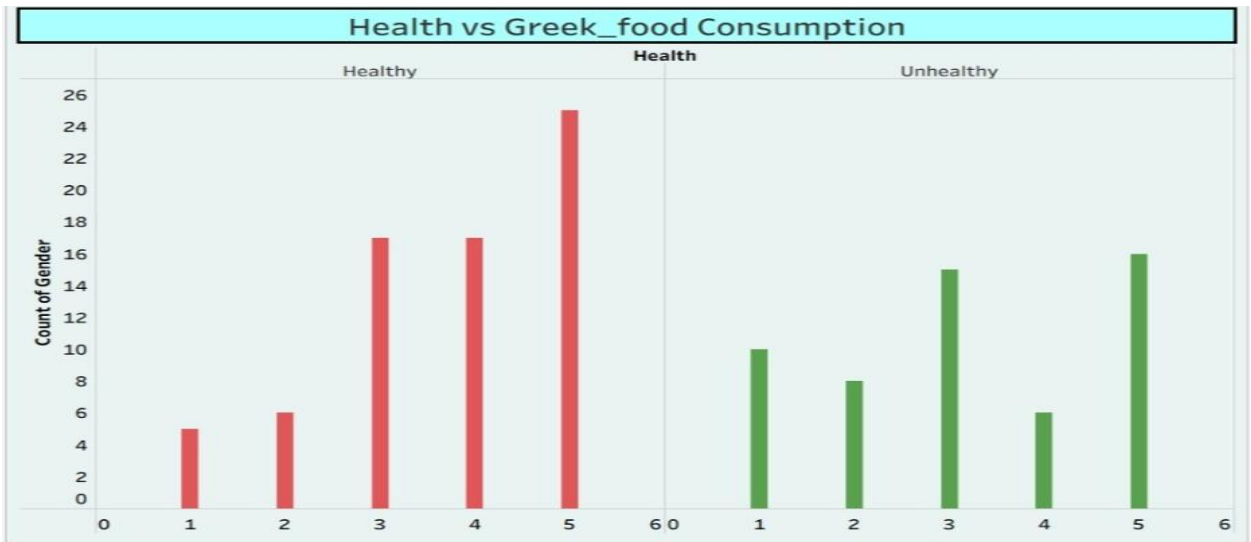
8. Data Verification:

Perform a final quality check to ensure the processed dataset is accurate, complete, consistent, and suitable for visualization. Verify that all transformations and calculations have been applied correctly.

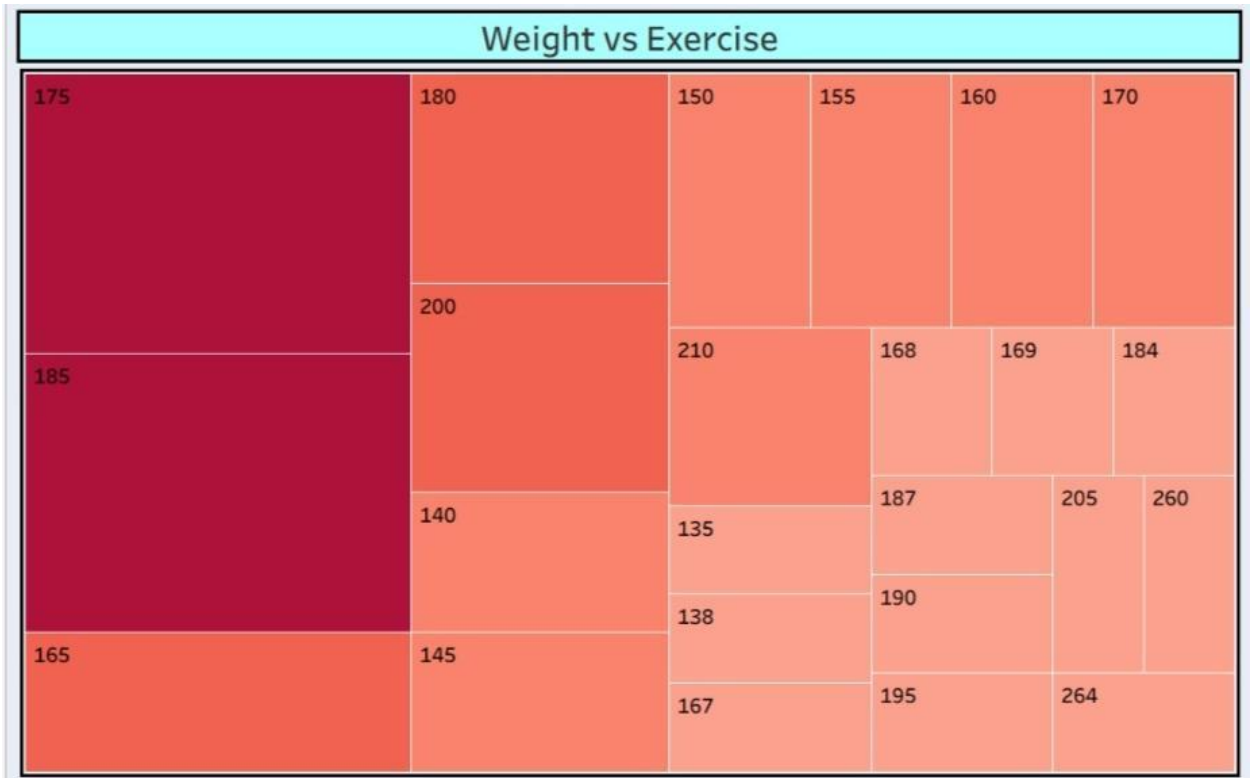
9. Data Visualization Preparation:

Load the cleaned dataset into Tableau, assign appropriate data types, create relationships if required, and develop interactive charts, dashboards, and visualizations to analyze college students' food choices, dietary patterns, nutritional habits, and health outcomes.

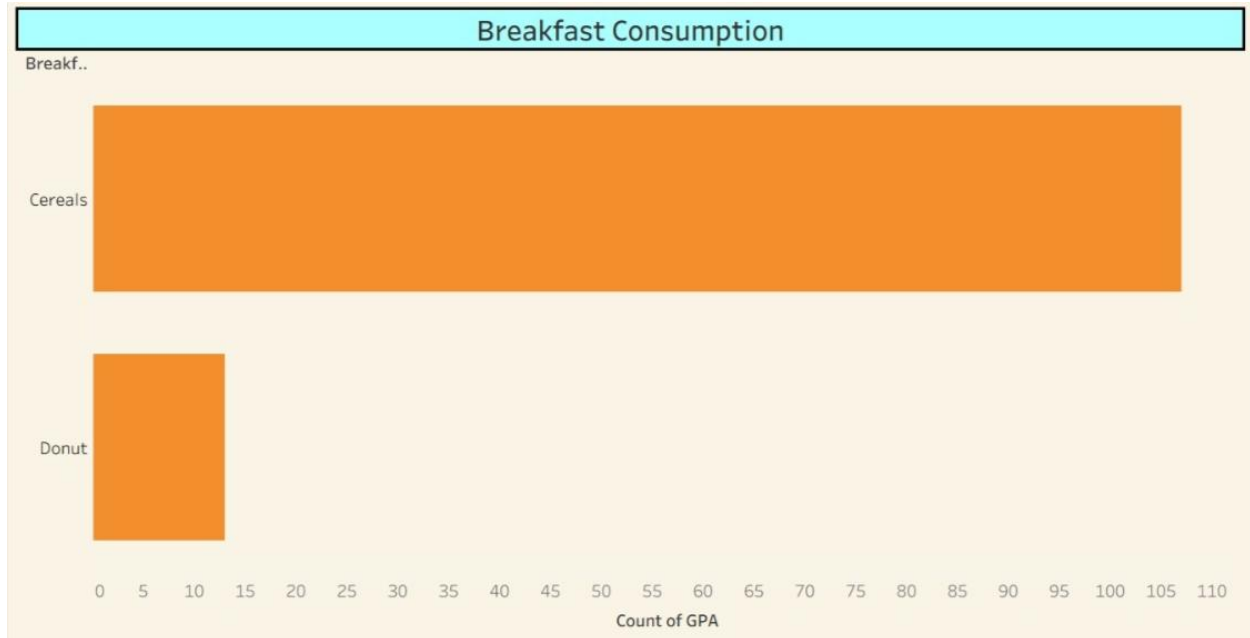
Business Question-1: How does Greek food consumption differ between healthy and unhealthy students, and what impact does it have on overall health status?



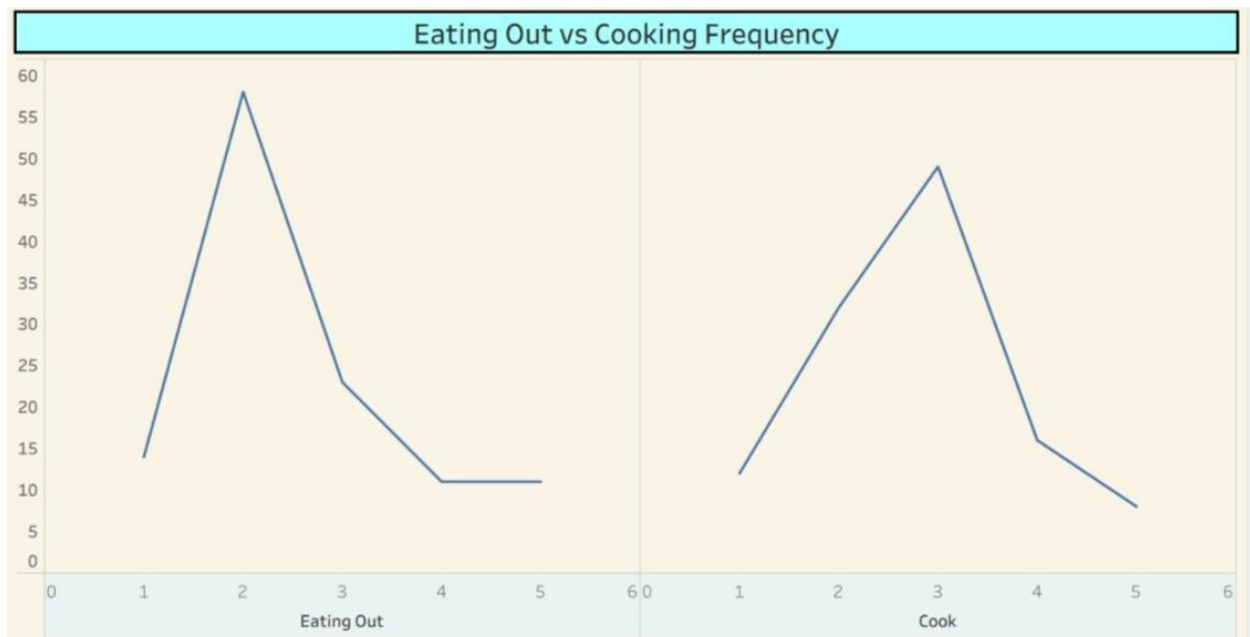
Business Question-2 : What is the relationship between students' weight and their exercise frequency, and how does physical activity influence weight management?



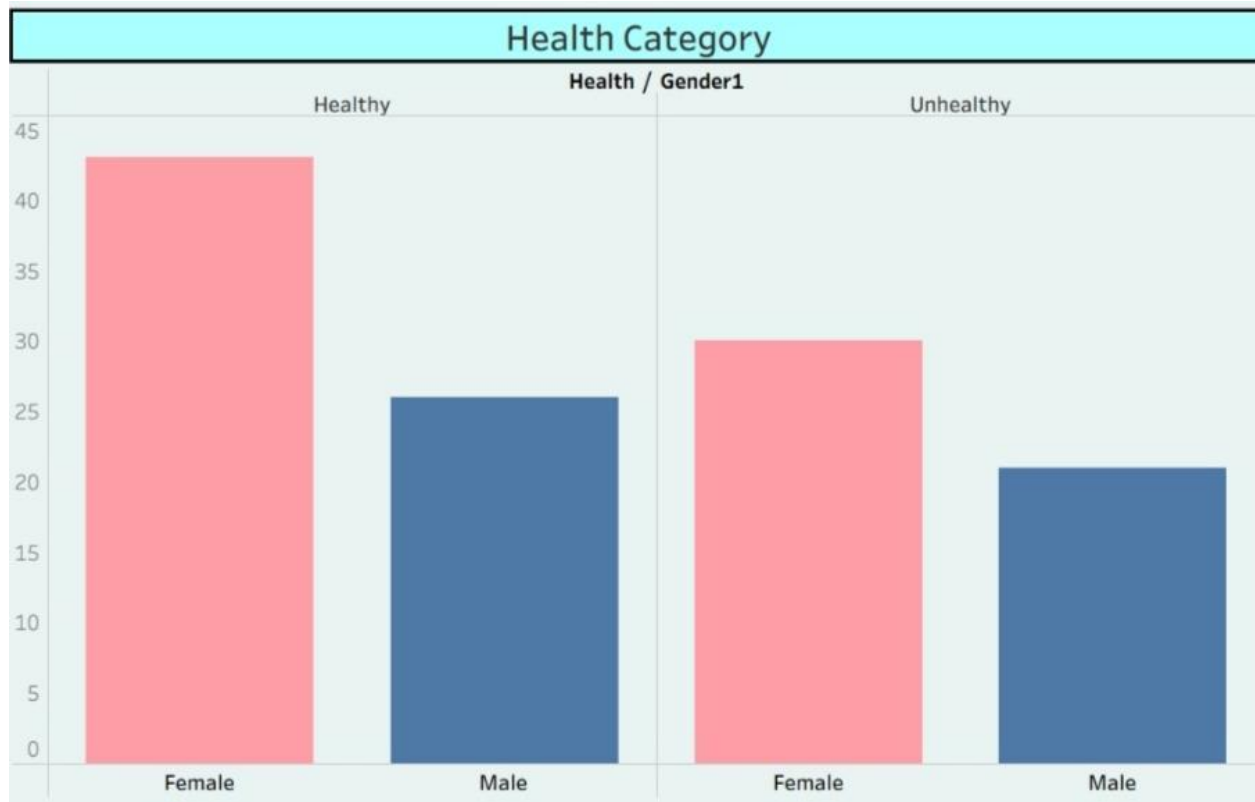
Business Question - 3 :Which breakfast option (e.g., cereals or donuts) is most commonly consumed by college students, and how does breakfast choice relate to academic and health outcomes?



Business Question - 4: How do eating-out habits compare with cooking frequency among students, and which habit is associated with healthier dietary patterns?



Business Question - 5: How does health status vary across male and female students, and which gender demonstrates healthier dietary behaviors?



Business Question - 6: What is the impact of vitamin consumption on students' overall health, and are students who consume vitamins more likely to maintain healthier lifestyles?

